

BENCHMARK: INACCURATE

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2456210.7005 +/- 0.0038 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.114 +/- 0.024                 |
| Transit depth (Rp/Rs)^2                           | 1.29 +/- 0.54 [%]               |
| Orbital Inclination (inc)                         | 89.48 +/- 1.45                  |
| Airmass coefficient 1 (a1)                        | 1.416 +/- 0.023                 |
| Airmass coefficient 2 (a2)                        | -0.34 +/- 0.15                  |
| Scatter in the residuals of the lightcurve fit is | 1.65 %                          |
| Best Comparison Star                              | #2 - [257, 298]                 |
| Optimal Aperture                                  | 4.25                            |
| Optimal Annulus                                   | 8.46                            |
| Transit Duration (day)                            | 0.0654 +/- 0.0042               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.114 ± 0.024 vs 0.1036 (deviation 0.41σ) |
| Duration fit vs published | 0.0654 d vs 0.1567 d (deviation 20.58σ)   |
| Mid-time O-C              | -7.9 min                                  |
| Depth SNR                 | 4.5                                       |
| Residual scatter          | 1.65 %                                    |

Light curve

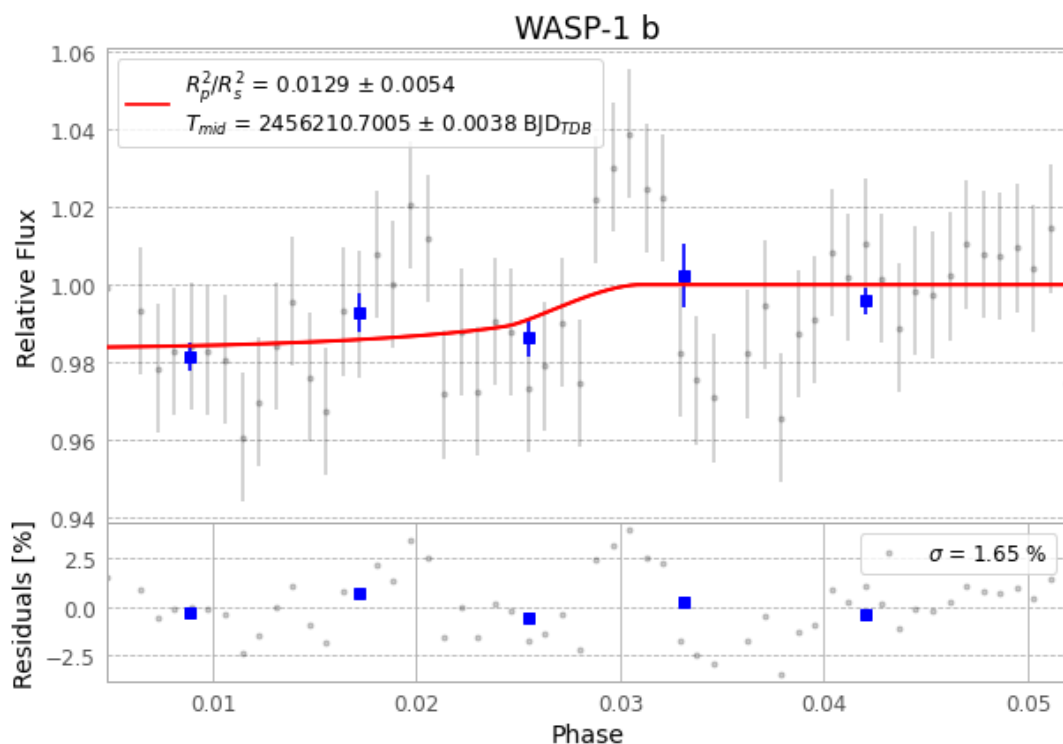


Figure 1 — FinalLightCurve\_WASP-1 b\_2012-10-09.png (SHA-256 a93227c838bd8504...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 244], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 4ceb642ef97273c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

58 FITS frames (38 MB), WASP-1121010045712.FITS → WASP-1121010074812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-121010.log (fa00422c18e4...), FinalLightCurve\_WASP-1 b\_2012-10-09.png (a93227c838bd...), FinalLightCurve\_WASP-1 b\_2012-10-09.csv (34602d84dfcc...)

## Compute resources

Wall time 154.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:19Z.